Consider any one angle among this say $\angle 1$. The angles which are adjacent to $\angle 1$ are $\angle 2$ and $\angle 4$, $\angle 3$ is a non-adjacent angle. Similarly, for the remaining three angles two angles will be adjacent and one angle will be non-adjacent. We can observe that an angle and its non-adjacent angle are just opposite to each other at the point of intersection $O$ (vertex). Such angles which are opposite to each other with reference to the vertex are called vertically opposite angles.

When two lines intersect each other, two pairs of non-adjacent angles formed are called vertically opposite angles.

## Try these

1. Four real life examples for vertically opposite angles are given below.

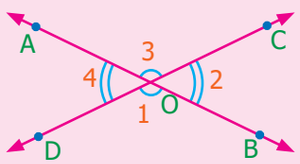




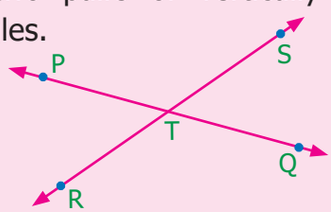

Give four more examples for vertically opposite angles in your surrounding.

2. In the given figure, two lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ intersect at $O$. Observe the pair of angles and complete the following table. One is done for you.

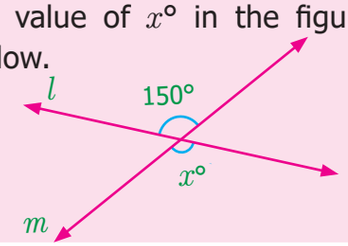
| Pair of angles | $\angle AOC$ | $\angle AOD$ | $\angle BOC$ | $\angle BOD$ |
|---|---|---|---|---|
| $\angle AOC$ | Same angle | Adjacent angle | Adjacent angle | Non – adjacent angle |
| $\angle AOD$ | | | | |
| $\angle BOC$ | | | | |
| $\angle BOD$ | | | | |




3. Name the two pairs of vertically opposite angles.




4. Find the value of $x°$ in the figure given below.




## Activity

On a paper draw two intersecting lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$. Let the two lines intersect at $O$. Label the two pairs of vertically opposite angles as $\angle 1$, $\angle 2$ and $\angle 3$, $\angle 4$. Make a trace of angles $\angle 2$ and $\angle 3$. Place the traced angle $\angle 2$ on angle $\angle 1$. Are they equal? Place the traced angle $\angle 3$ on angle $\angle 4$. Are they equal? Continue the same for five different pair of intersecting lines. Record your observations and discuss.



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